

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1510

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I _____ Shaik. Abdul Razak_(192524358)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____ Shaik. Abdul Razak _____

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

Problem:

A retail chain operates physical stores and an online platform. It generates large amounts of sales, inventory, and customer behaviour data. The company requires real-time inventory tracking, demand forecasting, and personalized marketing recommendations.

Proposed Big Data Architecture:

The architecture can follow the flow:

Data Sources → Data Ingestion → Data Storage → Stream/Batch Processing → Analytics & Machine Learning → Visualization & Applications

1. Data Sources:

- Point-of-sale (POS) systems from physical stores
- E-commerce transactions
- Inventory management systems
- Customer browsing and purchasing history
- Mobile applications and loyalty programs

2. Data Ingestion:

Apache Kafka can be used to collect real-time sales and inventory events. Apache NiFi can also be used for data collection and integration from different sources.

3.DataStorage:

A data lake using Hadoop HDFS or cloud object storage such as Amazon S3 can store large volumes of structured and unstructured data. A cloud data warehouse such as Amazon Redshift, Google Big Query, or Snowflake can be used for analytical queries.

4. Data Processing:

Apache Spark can process both batch and streaming data. Spark Streaming can continuously process sales and inventory events and update stock information in near real time.

5. Analytics and Machine Learning:

Demand forecasting can use time-series models such as ARIMA, Prophet, or LSTM. Customer segmentation can use clustering algorithms such as K-Means. Recommendation systems can use collaborative filtering and content-based filtering.

6. Visualization:

Power BI or Tableau can provide dashboards showing inventory levels, sales trends, demand forecasts, and customer behaviour.

Result:

The proposed architecture provides real-time inventory visibility, improves demand forecasting, reduces stock-outs and overstocking, and enables personalized product recommendations.

2. Social Media Fraud Detection Problem

Problem:

A social media platform processes billions of posts, comments, likes, and shares every day. It must detect fake accounts and misinformation quickly.

Proposed Big Data Architecture:

User Interactions → Kafka → Stream Processing → Feature Extraction → Machine Learning → Fraud/Misinformation Detection → Alert & Action

1. Data Sources:

- Posts and comments
- Likes and shares
- User profiles
- Login and activity patterns
- IP/device information
- Social connections

2. Data Ingestion:

Apache Kafka can collect billions of real-time events from the platform. Kafka provides scalable and fault-tolerant message processing.

3. Stream Processing:

Apache Flink or Spark Streaming can process events continuously. The system can calculate features such as posting frequency, unusual activity, account age, sharing patterns, and network behaviour.

4. Machine Learning:

Fake accounts can be detected using classification algorithms such as Random Forest, XGBoost, or neural networks. Anomaly detection can identify unusual behavior that differs from normal users.

Misinformation detection can use Natural Language Processing (NLP) models to analyze text, sentiment, keywords, and contextual information.

5. Storage:

Hadoop HDFS or cloud data lakes can store historical user activity. NoSQL databases such as Cassandra can provide scalable access to frequently required data.

6. Real-Time Response:

When suspicious activity is detected, the system can generate alerts, reduce content visibility, temporarily restrict accounts, or send the case for human verification.

Result:

The solution provides scalable and near-real-time detection of fake accounts and misinformation while supporting billions of daily social media interactions.

3. Government Data Platform Problem

Problem:

A government agency needs to integrate census, economic, and geospatial datasets from multiple departments. The platform must provide interoperability while maintaining strict privacy and security.

Proposed Governance and Data Management Strategy:

Data Sources → Data Integration → Central Data Lake → Data Governance → Secure Data Access → Analytics

1. Data Integration:

Data from different departments should be collected using standardized APIs and ETL/ELT pipelines. Apache NiFi or cloud integration services can support data ingestion.

2. Central Data Platform:

A government data lake can store census, economic, geographic, and administrative datasets. Structured data can also be stored in a data warehouse for reporting and analysis.

3. Data Standards and Interoperability:

Common data formats, metadata standards, APIs, and standardized identifiers should be used across departments. This allows different agencies to exchange and understand data consistently.

4. Data Governance:

A central governance framework should define:

- Data ownership
- Data quality standards
- Metadata management
- Data classification
- Data retention policies
- Data sharing rules

5. Security and Privacy:

Role-Based Access Control (RBAC), encryption, multi-factor authentication, network security, and audit logging should be implemented. Personally identifiable information should be anonymized or pseudonymized whenever possible.

6. Data Quality:

Automated validation should detect duplicate, missing, inconsistent, and incorrect records. A data catalog can help departments discover and understand available datasets.

7. Monitoring and Compliance:

All data access should be logged and monitored. Regular security audits and compliance checks should be performed.

Result:

The proposed governance strategy creates a secure and interoperable national data platform where government departments can share and analyze data while protecting citizen privacy.

4. Banking Fraud Detection Problem

Problem:

A bank processes millions of transactions every minute and must identify fraudulent transactions with very low latency and high accuracy.

Proposed Big Data Pipeline:

Transaction Sources → Kafka → Stream Processing → Feature Engineering → Fraud Detection Model → Decision Engine → Alert/Block → Data Lake

1. Data Sources:

- ATM transactions
- Credit and debit card transactions
- Online banking
- Mobile banking
- Historical transaction records
- Customer profiles and account information

2. Data Ingestion:

Apache Kafka can capture transactions as real-time streams. It provides high throughput and fault tolerance.

3. Stream Processing:

Apache Flink or Spark Streaming can process transactions in real time. Features such as transaction amount, location, time, transaction frequency, and unusual spending behavior can be calculated.

4. Machine Learning Algorithms:

Supervised algorithms such as Random Forest, XGBoost, and Logistic Regression can classify transactions as legitimate or fraudulent.

For previously unseen fraud patterns, anomaly detection algorithms such as Isolation Forest or Autoencoders can be used.

5. Historical Data:

Hadoop HDFS or cloud data lakes can store historical transactions. Historical data can be used to train and continuously improve fraud detection models.

6. Decision Engine:

The fraud score generated by the model can be used to make immediate decisions:

- Low risk → Approve transaction
- Medium risk → Request additional authentication

- High risk → Block transaction and generate alert

7. Monitoring:

Dashboards can monitor fraud rates, false positives, transaction volumes, and model performance.

Result:

The proposed pipeline provides low-latency fraud detection by combining real-time transaction streams with historical data and machine learning.

5. Agriculture Smart Farming Problem

Problem:

A smart farming system collects soil, weather, crop, drone, and sensor data from distributed locations. Farmers require real-time insights to improve crop yield and resource utilization.

Proposed Big Data Architecture:

Sensors/Drones/Satellites → IoT Gateway → Data Ingestion → Cloud Data Lake → Stream & Batch Processing → Analytics/ML → Farmer Dashboard

1. Data Sources:

- Soil moisture sensors
- Temperature and humidity sensors
- Weather stations
- Drone images
- Satellite imagery
- Crop growth information
- Irrigation and fertilizer data

2. Data Ingestion:

IoT gateways can collect sensor readings. MQTT can be used for lightweight communication between sensors and cloud services. Apache Kafka can handle high-volume event streams.

3. Data Storage:

A cloud data lake can store sensor readings, images, weather information, and historical agricultural data. Time-series databases can store continuously changing sensor values.

4. Data Processing:

Apache Spark or Apache Flink can process real-time sensor streams and historical agricultural datasets.

Image processing models can analyze drone and satellite images to identify crop stress, diseases, weeds, and changes in vegetation.

5. Analytics and Machine Learning:

Machine learning can be used for:

- Crop yield prediction
- Disease detection
- Irrigation prediction
- Soil condition analysis

- Weather-based farming recommendations
- Pest and weed detection

Regression models can predict crop yield, while classification and deep learning models can identify crop diseases from images.

6. Decision Support:

A farmer dashboard or mobile application can display soil moisture, weather conditions, crop health, irrigation requirements, and predicted yield.

For example, if soil moisture falls below a required level, the system can automatically recommend or trigger irrigation.

7. Benefits:

- Reduced water consumption
- Efficient fertilizer usage
- Early disease detection
- Improved crop yield
- Reduced operational costs
- Better resource management

Result:

The proposed big data solution combines IoT, cloud computing, stream processing, and machine learning to provide real-time agricultural insights and support precision farming decisions.

Overall Conclusion

Big data architectures can solve complex industry problems by combining scalable data storage, real-time stream processing, cloud platforms, machine learning, and data visualization. Retail, social media, government, banking, and agriculture systems can use these technologies to process large and diverse datasets, make faster decisions, improve efficiency, and provide better services.